Microprice as a Centered Poisson Corrector

Paper I: move-time structure, centering, and horizon effects

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Abstract. This paper has two goals. First, it identifies the object that classical microprice estimates after the event-time model is reduced to price-move time. For a finite top-of-book Markov model with no-move kernel Q, move kernel T, event-time one-step drift

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x],$$

and fundamental matrix $N = (I - Q)^{-1}$, define

$$G_1 := Nr, \qquad B := NT.$$

The n-move predictor is then

$$G^{(n)} = \sum_{j=0}^{n-1} B^j G_1.$$

Second, it explains why the long-run correction can fail at short horizons even when the move-time model is well specified. If the move-time chain is ergodic with stationary distribution π, mean move drift $\mu = \pi^\top G_1$, and projector $\Pi_B := \mathbf{1}\pi^\top$, then

$$G^{(n)} = n\mu\,\mathbf{1} + g - B^n g, \qquad (I - B)g = G_1 - \mu\mathbf{1}, \qquad \pi^\top g = 0,$$

and the long-run move-time microprice level is

$$G^* := (I - B + \Pi_B)^{-1} G_1 = g + \mu\mathbf{1}.$$

Thus classical microprice is a centered Poisson corrector on the move-time chain, while the transient term $-B^n g$ governs the horizon effect. We also study the move-time discounted resolvent $(I - \beta B)^{-1} G_1$, where β discounts across price moves after the reduction; this is distinct from the event-time discount γ used in the companion note *Microprice, Bellman Values, and Semi-Markov Reinforcement Learning*.

Empirically, this decomposition predicts a horizon effect: at short horizons the transient term can still dominate, so G_1 or even the raw mid can outperform G^*; at longer horizons the centered correction wins when the move-time martingale diagnostic remains small. Public BAC/CVX benchmarks, a five-name WSELOB panel, and a one-day IEX stress test all support that pattern. On WSELOB, the raw mid or G_1 is best at 10 seconds, whereas G^* or Disc(.99) is best on all five names by 60 seconds.

1 Introduction

Stoikov's microprice [1] estimates fair value from top-of-book state by propagating expected future price changes through an absorbing Markov chain. This paper is the structural half of a two-note sequence. Its aim is to identify the long-run object computed by the original construction and to explain the finite-horizon effects that appear when that long-run object is used as a short-horizon forecaster.

Throughout both notes we use the same basic notation. Let

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x], \qquad N := (I - Q)^{-1}, \qquad G_1 := Nr, \qquad B := NT.$$

If the move-time chain B is ergodic with stationary distribution π, we write

$$\mu := \pi^\top G_1, \qquad \Pi_B := \mathbf{1}\pi^\top, \qquad (I - B)g = G_1 - \mu\mathbf{1}, \qquad \pi^\top g = 0,$$

and

$$G^* := (I - B + \Pi_B)^{-1} G_1 = g + \mu\mathbf{1}.$$

Thus g is the centered move-time relative value, while G^* is the uncentered long-run move-time microprice level. In centered regimes $\mu = 0$, so the two coincide.

The division of labor between the two papers is deliberate. The present paper isolates the move-time Poisson structure, the martingale correction, and the empirical horizon crossover. The companion note keeps the same objects (r, N, G_1, B, g, G^*) and adds exact Bellman targets on event time, a semi-Markov interpretation on move time, and online TD-style learning rules. Read together, the first paper explains what microprice is asymptotically, while the second explains how the same construction fits into value-function language and horizon-aware forecasting.

The empirical section evaluates that prediction on three datasets. The public BAC/CVX benchmarks remain useful calibration checks, but they do not show uniform dominance: on BAC the weighted mid is still best under the grouped conditional-curve metric used here. The clearest positive evidence comes from WSELOB, where short- and long-horizon regimes separate cleanly. A one-day IEX venue-local panel then shows the complementary boundary case in which the centering diagnostic fails and discounted extrapolation becomes unreliable.

2 From event time to move time

Let X_n be a finite-state Markov approximation of the top of book at event time, and let p_n be the mid-price. Decompose the transition kernel according to whether the mid-price moves:

$$P = Q + T,$$

where Q_{xy} is the probability of moving from state x to state y without changing the mid-price and T_{xy} is the probability of moving from x to y with a price change. Define the event-time one-step drift

$$r(x) := \mathbb{E}[p_{n+1} - p_n \mid X_n = x].$$

Assume $\rho(Q) < 1$, so the no-move subchain is transient and

$$N := (I - Q)^{-1} = \sum_{k \geq 0} Q^k$$

exists. Set

$$G_1 := Nr, \qquad B := NT.$$

Proposition 2.1 (Embedded move-time chain)**.** *Let*

$$\tau_0 := 0, \qquad \tau_{k+1} := \inf\{n > \tau_k : p_n \neq p_{n-1}\}, \qquad Y_k := X_{\tau_k}.$$

Then Y_k is a Markov chain with transition kernel B. Moreover,

$$G_1(x) = \mathbb{E}_x[p_{\tau_1} - p_{\tau_0}],$$

so G_1 is the expected price change at the next move conditional on the current state.

The absorbing-chain recursion therefore reduces to a move-time Markov system: B controls transitions from one price move to the next, and G_1 is the corresponding one-move price drift. The same reduction also produces the event-time-discounted objects studied in the companion note: for $\gamma \in (0, 1)$,

$$G_{1,\gamma} := (I - \gamma Q)^{-1} r, \qquad B_\gamma := \gamma(I - \gamma Q)^{-1} T.$$

These are the move-time Bellman reward and continuation kernel for the option "wait until the next mid-price move." They discount each no-move excursion in event time before the excursion is collapsed. By contrast, the present paper later studies a move-time resolvent $(I - \beta B)^{-1} G_1$, where the discount is applied only after the reduction to price-move episodes. We therefore reserve γ for event-time discounting and β for move-time discounting.

If π denotes the stationary distribution of B, define

$$\mu := \pi^\top G_1, \qquad \Pi_B := \mathbf{1}\pi^\top.$$

Remark 2.2 (Centering)**.** *When B is ergodic, $B^n \to \Pi_B$. Hence Stoikov's convergence condition $B^n G_1 \to 0$ is equivalent to*

$$\pi^\top G_1 = 0.$$

Only the invariant constant mode can obstruct convergence.

3 The centered Poisson corrector

Define the n-move price adjustment by

$$G^{(n)} := \sum_{j=0}^{n-1} B^j G_1.$$

Equivalently,

$$\mathbb{E}[p_{\tau_n} - p_{\tau_0} \mid Y_0 = x] = G^{(n)}(x).$$

Proposition 3.1 (Finite-horizon decomposition)**.** *Assume that B is ergodic with stationary distribution π, and set*

$$h := G_1 - \mu\mathbf{1}.$$

Let g be the unique centered solution of

$$(I - B)g = h, \qquad \pi^\top g = 0.$$

Then, for every $n \geq 1$,

$$G^{(n)} = n\mu\mathbf{1} + g - B^n g.$$

If

$$G^* := (I - B + \Pi_B)^{-1} G_1,$$

then

$$G^* = g + \mu\mathbf{1}.$$

In particular, if $\mu = 0$, then $G^* = g$ and

$$G^{(n)} \to G^*.$$

The predictor therefore splits into three pieces: a linear drift term $n\mu\mathbf{1}$, a bounded state correction g, and a transient remainder $-B^n g$. The centered microprice is the bounded component. The empirical horizon effect comes from the transient term: at short horizons it can still dominate the limiting correction.

Corollary 3.2 (Move-time martingale correction)**.** *Under the assumptions of Proposition 3.1,*

$$M_k := p_{\tau_k} - k\mu + g(Y_k)$$

is a martingale with respect to the move-time filtration. When $\mu = 0$, the corrected price

$$p_{\tau_k} + G^*(Y_k)$$

is a move-time martingale.

Proposition 3.3 (Move-time discounted regularization)**.** *For $\beta \in (0, 1)$ define*

$$g_\beta := (I - \beta B)^{-1} G_1.$$

Then

$$g_\beta = \frac{\mu}{1-\beta}\mathbf{1} + (I - \beta B)^{-1}(G_1 - \mu\mathbf{1}).$$

Hence the only singular term as $\beta \uparrow 1$ is the constant mode. If $\mu = 0$ and, for some $\rho \in (0, 1)$,

$$\|B^k G_1\|_\infty \le C_\rho \rho^k \|G_1\|_\infty \qquad \text{for all } k \ge 0,$$

then

$$\|g_\beta - G^*\|_\infty \le (1-\beta)\frac{C_\rho \rho}{(1-\rho)^2}\|G_1\|_\infty.$$

The resolvent is always well posed, even when sample centering is imperfect. In centered regimes it is nearly lossless; in drifting or weakly centered regimes its constant mode can dominate. It is therefore best viewed as a regularized proxy and a useful stress test, not as an automatic substitute for the centered correction. It also differs conceptually from the discounted Bellman values in the companion note: $(I - \beta B)^{-1} G_1$ discounts after the reduction to move time, whereas $V_\gamma = (I - \gamma P)^{-1} r$ discounts before the reduction and therefore reweights no-move excursion lengths.

4 Empirical results

4.1 Evaluation protocol

We evaluate predictors against grouped state-conditioned future-price curves rather than pointwise forecast errors. For a horizon h, let c index a spread/imbalance bucket. Write $\bar{\Delta}_h(c)$ for the average future mid-price change over all rows in bucket c, and write $\bar{f}(c)$ for the average predictor value in the same bucket. The reported score is

$$\mathrm{RMSE}_h(f) := \left(\frac{1}{|C_h|} \sum_{c \in C_h} \left(\bar{\Delta}_h(c) - \bar{f}(c) \right)^2 \right)^{1/2}.$$

The horizons are fixed clock/event-time horizons rather than exact move counts, so the empirical question is qualitative: does the long-run correction become more useful as the horizon lengthens? Throughout this paper, "Disc(.99)" means the move-time resolvent $(I - 0.99\,B)^{-1} G_1$; the companion note reserves $V_{0.99}$ for event-time discounting on P.

To connect the data back to Corollary 3.2, we also report the move-time martingale diagnostic. Writing

$$\Delta P_k := p_{\tau_{k+1}} - p_{\tau_k}, \qquad \widehat{\xi}_k := \Delta P_k - \widehat{\mu} + G^*(Y_{k+1}) - G^*(Y_k), \qquad \widehat{t}_M := \left| \frac{\overline{\widehat{\xi}}}{s_{\widehat{\xi}}/\sqrt{n}} \right|.$$

Small values of $\widehat{t}_M$ indicate that the estimated correction is close to a move-time martingale.

4.2 Public BAC/CVX benchmarks

Table 1 and Figure 1 summarize the public BAC/CVX benchmarks. On BAC, G^* reduces grouped RMSE by about 51% at 60 seconds and 76% at 180 seconds relative to the raw mid, but the weighted mid remains best on this metric. On CVX, by contrast, G^* reduces grouped RMSE by roughly 62% and decisively beats the weighted mid. In all three benchmark cases, $\widehat{t}_M < 2$.

Table 1. Public BAC/CVX grouped-RMSE benchmarks and move-time diagnostics.

Asset	Horizon (s)	Raw mid	Weighted mid	G_1	G^*	Disc(.99)	$\widehat{t}_M$
BAC	60	0.001812	0.000435	0.000909	0.000892	0.000892	1.12
BAC	180	0.002287	0.000345	0.000561	0.000547	0.000547	1.12
CVX	10	0.001828	0.003941	0.000910	0.000697	0.000699	0.84

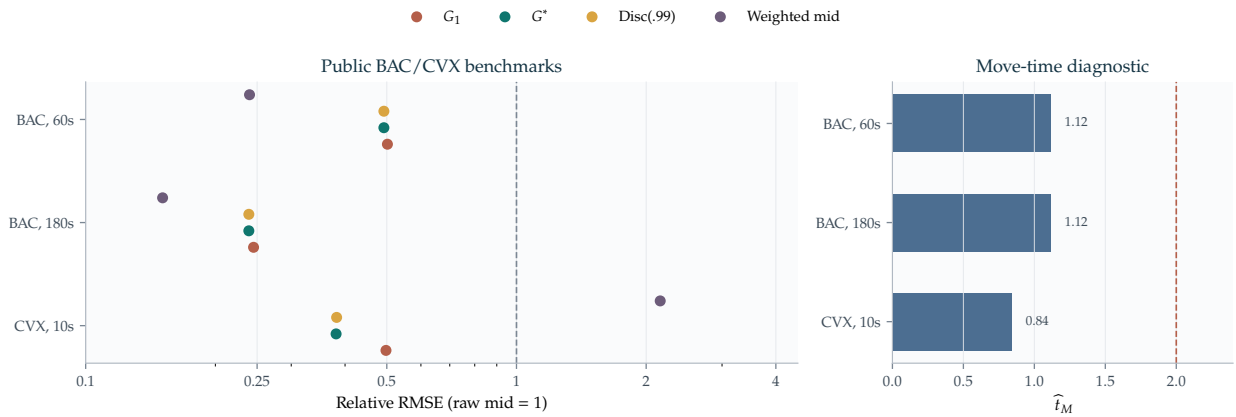

Figure 1. Public BAC/CVX benchmarks. Left: grouped RMSE relative to the raw mid. Right: the move-time martingale diagnostic. BAC remains a nuanced benchmark in which the weighted mid is highly competitive, whereas CVX is the cleaner public success case for G^*.

4.3 WSELOB: the horizon crossover

WSELOB gives the clearest horizon crossover. At 10 seconds, the raw mid is best on PEKAO, PKOBP, and PZU, while G_1 is best on KGHM and PKNORLEN. On those same names, G^* is still between 6% and 83% worse than the raw mid, which is exactly what one expects when the transient term $-B^n g$ remains material.

By 60 seconds the ordering reverses. G^* is best on four names and Disc(.99) is best on the fifth. Relative to the raw mid, G^* reduces grouped RMSE by between 20% and 56% on all five names. The weighted mid is not competitive: it is between 4.7× and 11.0× the raw-mid error at 10 seconds, and between 2.1× and 6.6× the G^* error at 60 seconds. The move-time diagnostic remains comfortably in the centered regime, with $\widehat{t}_M < 2$ for every name.

Table 2. WSELOB summary by asset.

Asset	Best @ 10s	Best @ 60s	Raw/G^* @ 60s	WMid/G^* @ 60s	$\widehat{t}_M$
KGHM	G_1	G^*	1.25	2.07	1.97
PEKAO	Raw mid	Disc(.99)	1.97	6.61	1.08
PKNORLEN	G_1	G^*	1.54	4.14	0.68
PKOBP	Raw mid	G^*	2.03	4.94	1.85
PZU	Raw mid	G^*	2.26	6.37	0.77

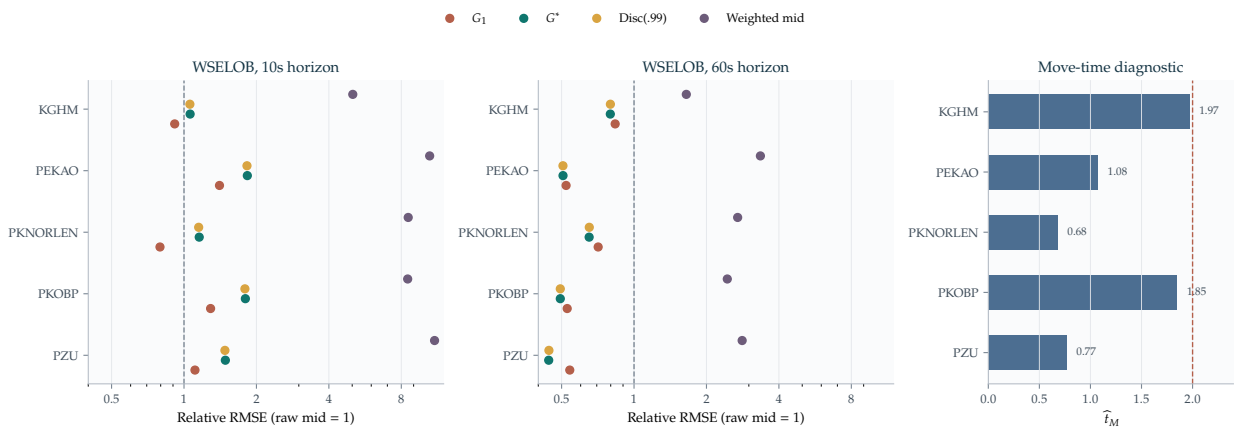

Figure 2. WSELOB summary. Relative RMSE is normalized by the raw mid, so values below one indicate an improvement over the raw mid. The crossover is clear: the limiting correction is not yet best at 10 seconds, but it becomes best by 60 seconds once the horizon is long enough.

4.4 IEX: centering as a diagnostic

The IEX venue-local panel is deliberately harder: it uses a single venue, a short sample, and a coarse state description, so residual drift is more likely. BAC remains near-centered with $\widehat{t}_M = 0.16$, and G^* is modestly better than the raw or weighted mid at both horizons. SPY and GLD look different. Their diagnostics are $\widehat{t}_M = 3.09$ and 2.24, respectively, and the discounted surrogate no longer shadows the centered correction. At 10 seconds, Disc(.99) is about 3.16× the G^* error on GLD and 9.50× the G^* error on SPY. This is exactly the failure mode predicted by Proposition 3.3: once the constant mode is not well centered, discounted extrapolation can be dominated by residual drift rather than by the bounded state correction.

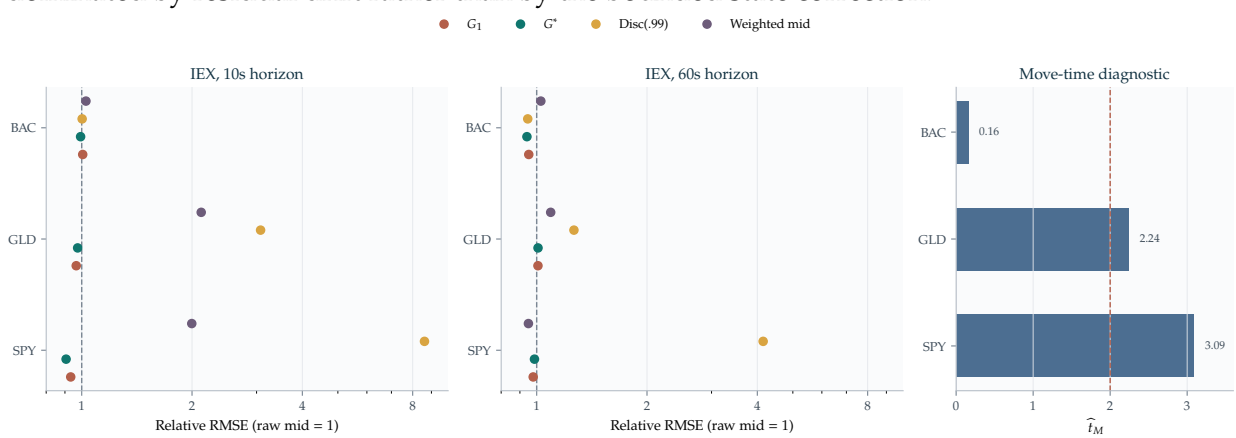

Figure 3. IEX venue-local stress test. BAC remains near-centered, while SPY and GLD violate the move-time martingale diagnostic. In those names the discounted resolvent no longer behaves like a harmless regularization of G^*.

5 Conclusion

Sampling at move times turns the microprice construction into a finite-state Poisson problem. The event-time recursion collapses to a move-time kernel B and a one-move drift G_1, while the n-move predictor admits the exact decomposition

$$G^{(n)} = n\mu\mathbf{1} + g - B^n g.$$

This representation clarifies both the martingale correction and the empirical horizon dependence of the estimator.

Across datasets, the centered correction is most useful when two conditions hold at once: the horizon is long enough for the transient term to decay, and the move-time martingale diagnostic remains small. WSELOB is the clearest positive case. BAC/CVX confirm the public benchmark, with the important qualification that BAC still favors the weighted mid under the grouped metric used here. The IEX panel shows the complementary failure mode: when centering weakens, discounted regularization amplifies the wrong mode rather than recovering the bounded state correction.

The companion RL note starts from exactly this dictionary. It keeps (r, N, G_1, B, g, G^*) fixed, replaces the move-time regularizer by exact event-time Bellman values $V^{(H)}$ and V_γ, and shows that the same price-move episodes support both differential TD for the classical centered correction and semi-Markov TD for discounted forecasting.

Read together, the two papers separate the asymptotic and the objective-matched questions cleanly. The present paper explains the move-time Poisson structure, the martingale diagnostic, and the horizon crossover of G^*. The companion note explains how the same reduced model fits into Bellman, average-reward, and semi-Markov RL language without changing the underlying microprice primitives.

A Proofs

Proof of Proposition 2.1. The first price move occurs after n no-move steps and one move step with kernel Q^nT. Summing over $n \geq 0$ gives

$$\mathbb{P}_x(Y_1 = y) = \sum_{n\geq 0}(Q^nT)_{xy} = ((I-Q)^{-1}T)_{xy} = B_{xy}.$$

Thus B is the move-time transition kernel. For the reward identity,

$$\mathbb{E}_x[p_{\tau_1} - p_{\tau_0}] = \sum_{n\geq 0}(Q^n r)_x = ((I-Q)^{-1}r)_x = G_1(x).$$

The Markov property of (Y_k) follows from the strong Markov property at the stopping times (τ_k). □

Proof of Proposition 3.1. Write $G_1 = \mu\mathbf{1} + h$ with $\pi^\top h = 0$. Since $(I-B)g = h$,

$$\sum_{j=0}^{n-1} B^j h = \sum_{j=0}^{n-1} B^j(I-B)g = \sum_{j=0}^{n-1}(B^j g - B^{j+1}g) = g - B^n g.$$

Therefore

$$G^{(n)} = \sum_{j=0}^{n-1} B^j G_1 = n\mu\mathbf{1} + g - B^n g.$$

Next,

$$(I - B + \Pi_B)(g + \mu\mathbf{1}) = (I - B)g + \Pi_B g + \mu\Pi_B\mathbf{1} = h + 0 + \mu\mathbf{1} = G_1,$$

so $G^* = g + \mu\mathbf{1}$. If $\mu = 0$, then $G^* = g$ and ergodicity gives

$$B^n g \to \Pi_B g = \mathbf{1}\pi^\top g = 0,$$

hence $G^{(n)} \to G^*$. Uniqueness of g under $\pi^\top g = 0$ follows because the nullspace of $I - B$ consists of constants. □

Proof of Corollary 3.2. By Proposition 3.1,

$$(I - B)g = G_1 - \mu\mathbf{1}.$$

Hence

$$\mathbb{E}[p_{\tau_{k+1}} - p_{\tau_k} \mid Y_k] = G_1(Y_k),$$

while

$$\mathbb{E}[g(Y_{k+1}) - g(Y_k) \mid Y_k] = (Bg - g)(Y_k) = -(G_1(Y_k) - \mu).$$

Adding the two conditional expectations yields

$$\mathbb{E}[M_{k+1} - M_k \mid Y_k] = 0.$$

□

Proof of Proposition 3.3. Because $\|\beta B\|_\infty \le \beta < 1$, the Neumann series converges and

$$g_\beta = \sum_{k\ge 0} \beta^k B^k G_1.$$

Using $G_1 = \mu\mathbf{1} + h$ and $B^k\mathbf{1} = \mathbf{1}$,

$$g_\beta = \sum_{k\ge 0} \beta^k(\mu\mathbf{1} + B^k h) = \frac{\mu}{1-\beta}\mathbf{1} + (I - \beta B)^{-1}h.$$

If $\mu = 0$, then $h = G_1$ and

$$G^* - g_\beta = \sum_{k\ge 0}(1 - \beta^k)B^k G_1.$$

The $k = 0$ term vanishes. For $k \ge 1$, $1 - \beta^k \le k(1 - \beta)$, so

$$\|G^* - g_\beta\|_\infty \le (1 - \beta)\sum_{k\ge 1} k\,\|B^k G_1\|_\infty \le (1 - \beta)C_\rho\|G_1\|_\infty \sum_{k\ge 1} k\rho^k.$$

Since $\sum_{k\ge 1} k\rho^k = \rho/(1 - \rho)^2$, the stated bound follows. □

References

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